

CHARACTERISTICS OF SEMICONDUCTOR GAS SENSORS

I. STEADY STATE GAS RESPONSE

P K CLIFFORD and D T TUMA

Electrical Engineering Department, Carnegie-Mellon University, Pittsburgh, PA 15213 (U S A)

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Abstract

The steady state gas sensing characteristics of SnO_2 gas sensors, as exemplified by the Taguchi Gas Sensor (TGS), are comprehensively studied. Resistance responses to hydrogen, methane, carbon monoxide and water vapor are experimentally characterized, with particular emphasis on multiple gas interactions. We find that the presence of ambient oxygen is essential to the sensor's operation and that the detection of combustible or reducing gas is mediated by reaction with adsorbed oxygen on the sensor surface.

A quantitative model of device operation is constructed which unifies the diverse properties of the TGS and other semiconductor sensors. The model supplies the mathematical framework for meaningful comparisons of sensor performances and gas sensitivities. In addition, it provides for the intrinsic power law response of sensor electrical resistance to combustible gas concentrations, the competitive and synergistic interactions of several gases detected simultaneously, and the source of sensor unselectivity. Stoichiometries and activation energies for the pertinent oxidation reactions on the tin oxide sensor surface can be derived from a sensor's resistance response by curve fitting to the mathematical model.

Key words: tin oxide, gas detection, oxygen adsorption, carbon monoxide semiconductor sensors, chemical sensing

Introduction

Semiconductor gas sensors offer the potential for developing portable, rugged and inexpensive gas detection instrumentation. However, this potential has not been developed. Although present sensors respond to the presence of ambient combustible gases in low ppm concentrations, their non-linearity, irreproducibility, drift and very poor selectivity preclude their use in sensitive and selective gas analysis. Such undesirable behavior will be rectified only

when knowledge of the basic physical mechanisms of gas detection is acquired

To acquire quantitative knowledge of the gas detection process in semiconductor gas sensors, we studied a specific commercially available tin dioxide sensor, the Taguchi Gas Sensor (TGS) manufactured by Figaro Inc of Japan. It was chosen for its great sensitivity to flammable gases, its availability and its reliable production. Corollary studies [1] of ZnO and SnO₂ sputtered films indicate that the TGS behavior is characteristic of state-of-the-art semiconductor sensors.

Previous studies of the TGS response [2 - 4], while providing some experimental data, were insufficiently quantitative to lead to a comprehensive model of the response, or an understanding of the physical mechanisms involved in its operation. Indeed, the most salient feature of the response, its power law nature, has not been described sufficiently to determine its physical basis. Consequently, the objective of the present investigation has been to acquire a detailed quantitative description of the diverse attributes of the TGS. Against this empirical understanding new models of device behavior are critically tested.

In this investigation two quantitatively controlled techniques are pursued in an effort to elicit the basic mechanisms of sorption, surface catalysis, electrical interaction and detection of gases by the TGS. The first technique is reported in this paper. It involves the measurement of sensor resistance response to combustible gases in air ambients at fixed sensor temperature. The second technique consists of the measurement of the dynamic response of sensor conductance to step temperature changes at different fixed oxygen gas concentrations in a nitrogen ambient. These dynamic measurements supply data regarding the kinetics of sorption and the mechanisms of charge exchange between the semiconductor volume and adsorbant surface states, and are reported in a companion paper.

The investigation of the TGS static gas response, reported here, clarified the role of oxygen in the response to combustible (reducing) gases, determined the causes of the poor selectivity of the TGS, quantified the response to various gases and provided detailed understanding of the interactions between various gases. These results relied on precise measurement of the TGS gas response to combinations of oxygen, nitrogen, hydrogen, methane, carbon monoxide and water vapor. In particular, the previously noted [5] interaction of carbon monoxide and water vapor on SnO₂ surfaces is thoroughly quantified. The power law nature of the response is mathematically modeled and the physical plausibility of the model is discussed. A more detailed theoretical model will be presented in a future paper.

After presenting the experimental technique, representative results are reported accompanied by a brief discussion of their notable aspects. The over-all experimental results and their relation to previous findings are then discussed. The conclusion summarizes the importance of this work and its impact on semiconductor gas detection technology.

Experimental procedure

A bank of ten TGS #812 is mounted on a Teflon board in a nine liter glass bottle. Gas mixtures are introduced to the bottle at atmospheric pressure from a gas mixing system capable of reliable production of one ppm to 10^6 ppm volumetric concentrations of test gases in a carrier gas. All gases are of ultra high purity (uhp) grade, containing less than 3 ppm of contaminants. The ambient inside the glass bottle is continuously mixed by a magnetically coupled Teflon stirrer. Two ambient gases are generally used: 100% O_2 and constituted dry air (79% N_2 and 21% O_2).

The TGS is composed of a thin layer of sintered SnO_2 material on a small (4 mm $\times$ 1 mm) ceramic cylinder. Two electrodes contact the ohmic semiconductor layer, enabling resistance measurement. A small coil of wire within the ceramic support acts as a heater. A data acquisition, reduction and analysis system consisting of two microcomputers is used. One monitors the TGS resistances ($\pm 2.5\%$ precision, $10\ \Omega$ to $10^6\ \Omega$) and controls their operating temperature, the other stores the data on floppy disks and, in turn, reduces the data and fits it to mathematical models.

Most gas response measurements are performed with the sensors operated at a constant sensor heating coil voltage of 5 V, yielding a surface temperature of 718 ± 5 K in air. The implications of sensor operating temperature are investigated in a separate experiment in which the TGS response to methane is measured at various temperatures. Details of the experimental set-up and procedure can be found elsewhere [1].

Results and analyses

Nitrogen and argon

In all measurements argon replaces nitrogen as ambient gas without affecting sensor resistance. When combinations of nitrogen and oxygen are used, argon and oxygen may be used without any differences in results. It is concluded that both nitrogen and argon are inert with respect to the TGS behavior.

Oxygen

The measurement of the TGS response to oxygen is complicated by a behavior which does not appear in the measurement of the response to low concentrations of reducing gases in an oxygen or air ambient. In the latter case, there is a rapid (< 2 min) response of TGS resistance to an injection of reducing gas followed by a long period of stable resistance. In the former, a reduction of oxygen pressure from near atmospheric to a few ppm levels results in a similar rapid response, but this is followed by a slow but significant continuous drift in resistance. To reduce the effects of this drift, the experiment is performed in times short compared to the drift time.

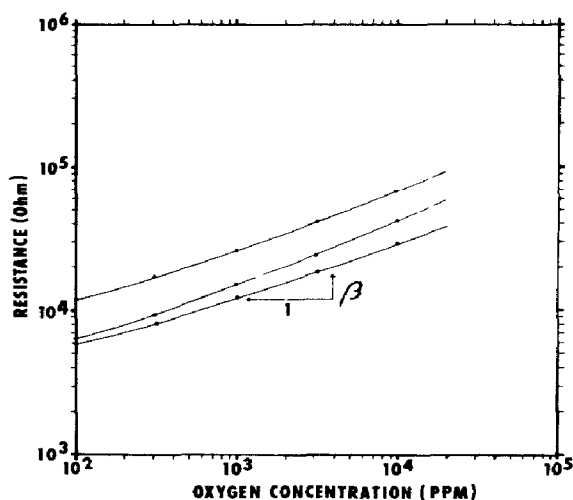


Fig 1 Resistance response of several TGS to oxygen. The ambient gas is nitrogen. At high concentrations a power law is evident with exponent β .

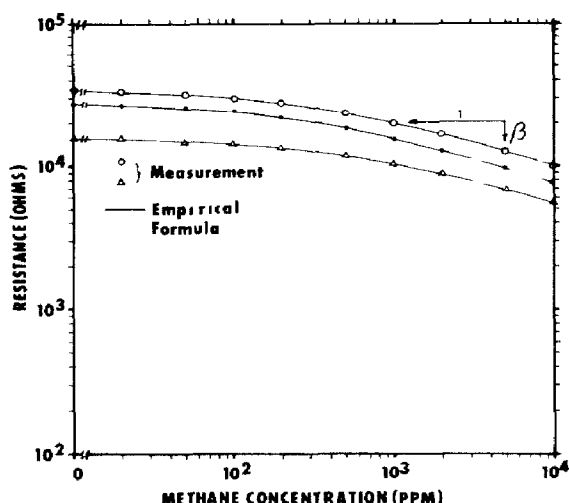


Fig 2 Resistance response of several TGS to methane in air. The solid lines represent curve fits of the experimental points to eqn (2).

Representative measurements are shown in Fig 1. The resistances are plotted against oxygen concentrations for three of the ten sensors. These log-log plots approach straight lines for high concentrations, indicating the power law nature of the response,

$$R = R_0 P_{O_2}^\beta \quad (1)$$

where R_0 is the sensor resistance in air, P_{O_2} is the relative oxygen partial pressure (P_{O_2} takes the value one for air and five for pure oxygen at atmospheric pressure), and β is the power law exponent, which varies from one detector to another from a minimum of about 0.25 to a maximum of about 0.55, and corresponds to the slope of the plots of Fig 1.

Methane

The same ten sensors, whose typical oxygen response is shown in Fig 1, are allowed to stabilize in air. Their responses to concentrations of methane extending to 10 000 ppm in air are measured. The resistance responses of the two sensors of extreme resistance are plotted with that of one which typifies the mean in Fig 2. The significant features of this response are an asymptotic approach to constant resistances at very low concentrations and an asymptotic approach to a power law relationship at high concentrations. A simple mathematical expression which describes the essentials of the response curve is the following

$$R = R_0 (1 + K_{CH_4} [CH_4])^{-\beta} \quad (2)$$

where $[\text{CH}_4]$ is the concentration of methane expressed in volumetric ppm at atmospheric pressure, and K_{CH_4} is a constant of dimension ppm^{-1} . This expression is curve fitted to the measured resistances to produce the solid curves of Fig. 2. The average sample standard deviation of the measurements from the fits of eqn. (2) is approximately equal to the measurement error, $\pm 2\%$. For the middle curve of Fig. 2, R_0 , K_{CH_4} and β are determined to be $27.4 \text{ k}\Omega$, $4.37 \times 10^{-3} \text{ ppm}^{-1}$ and 0.34 , respectively. The resistance in clean air, R_0 , varies widely from one detector to another. The values of K_{CH_4} show a similar variation. The power law slope, β , is characteristic of the particular sensor. For a collection of ten sensors, β shows the same variance it exhibits in the oxygen response, values range from 0.25 to 0.55 .

For a particular sensor the power law slope is of the same magnitude for methane as it is for oxygen, but it is of opposite sign. For this reason the methane response is termed 'first order with respect to oxygen' or just 'first order'. In a later experiment it is shown that the TGS resistance response to some gases, particularly hydrogen and sometimes carbon monoxide, is characterized by a power law slope which is twice that of the response to oxygen. For those cases we adopt the terminology 'second order'. By zero-order response we indicate a constant resistance independent of contaminant gas pressure or concentration. We define the threshold of detection as the methane concentration at which the response switches from zero order to first order. This occurs when $[\text{CH}_4] = K_{\text{CH}_4}^{-1}$. It is the position of this transition between zero- and first-order response which identifies the parameter K_{CH_4} as the sensitivity coefficient for methane, and largely determines the sensor usefulness as a gas detector.

The response of a single TGS to methane in a dry air ambient is contrasted to the response in a pure oxygen ambient in Fig. 3. A perusal of eqn. (1) and eqn. (2) suggests that an oxygen pressure different from air concentration may affect the response as

$$R = R_0 P_{\text{O}_2}^\beta (1 + K_{\text{CH}_4} [\text{CH}_4])^{-\beta} \quad (3)$$

Here, we again adopt the convention of defining the partial pressure of oxygen in air to be one. This enables R_0 to have the meaning 'resistance in air' even though oxygen pressures other than that of air may be used.

The fit of eqn. (3) to the experimental data, shown in Fig. 3, constrains β to the value of 0.52 , previously determined for this sensor. With this constraint, the computed sensitivity coefficients for the two curves are close 0.011 ppm^{-1} for the oxygen ambient and 0.009 ppm^{-1} for the air ambient. The differences between the two curves, evident in the uniform vertical translation, can be attributed solely to the term in oxygen pressure, $P_{\text{O}_2}^\beta$.

The form of eqn. (3) suggests that the sensor resistance at high methane concentrations is determined by the ratio of methane to oxygen concentrations

$$R^{1/\beta} \sim [\text{O}_2]/[\text{CH}_4]$$

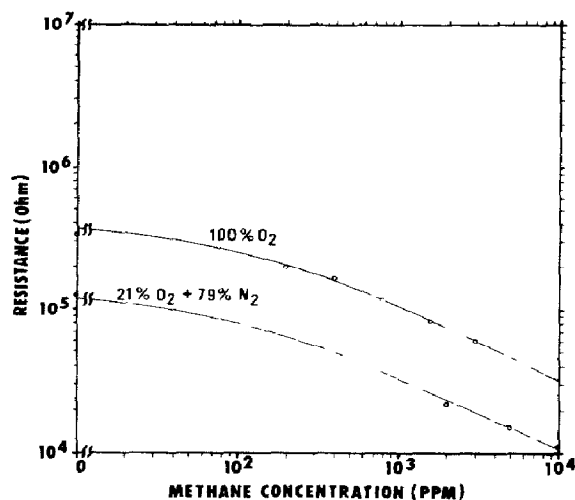


Fig 3 Resistance response of a single sensor to methane for ambients of different oxygen content at atmospheric pressure. The solid lines represent fits of each set of measurements to eqn (3). The difference between the two curves results primarily from the oxygen pressure term of eqn (3).

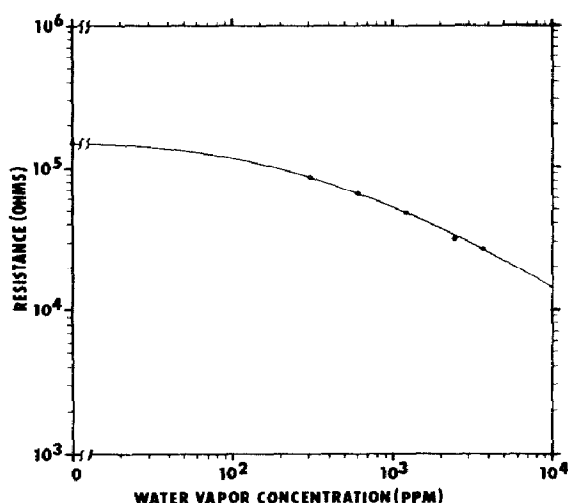


Fig 4 Resistance response of the TGS to water vapor in air. The solid line represents eqn (4).

It is this suggestion of a mass action law that justifies referring to the methane response as 'first order with respect to oxygen'.

Water vapor

Gas sensors used in typical applications can be exposed to a wide range of humidity in air. Consequently, the effects of water adsorption on the sensor surface are of great importance. The TGS manufacturer provides some scant data concerning the TGS response to water vapor in the form of graphs for 'correcting' the hydrocarbon response for different ambient conditions of humidity or temperature. To our knowledge there have been no systematic studies of the effect of H_2O adsorption on the TGS other than the qualitative observations of Boyle and Jones [2].

The TGS response to water vapor for a single sensor, Fig 4, is fitted to the equation

$$R = R_0(1 + K_{H_2O}[H_2O])^{-\beta} \quad (4)$$

with $R_0 = 151 \text{ k}\Omega$, $K_{H_2O} = 0.0059 \text{ ppm}^{-1}$ and $\beta = 0.55$. The water vapor response is first order with respect to oxygen. Again, the points of Fig 4 represent measurements and the solid curve represents a curve fit of eqn (4).

The sensor sensitivity to water vapor is somewhat less than that to methane. However, the ubiquitous nature of water vapor elevates the response to one of great practical importance. A consequence of the high ($\sim 10^4 \text{ ppm}$) water vapor concentration in room air is that the detection of any other reducing gas is expected to differ from what it would be for a dry ambient.

Hydrogen

The measured response of the TGS resistance to hydrogen gas is illustrated in Fig 5 for two atmospheres, the first 100% oxygen, and the second air. Just as with the methane response, the differing oxygen concentrations in the ambient cause a uniform vertical translation of the entire response, that is,

$$R \sim R_0 P_{O_2}^\beta$$

The power law response evident in Fig 5 is fitted well with an equation similar to that for the methane and water responses

$$R = R_0 P_{O_2}^\beta (K_0 + K_{H_2} [H_2]^2)^{-\beta} \quad (5)$$

with an average $K_{H_2} = 0.12 \text{ ppm}^{-2}$. Independent determinations of the sensitivity coefficients, K_{H_2} , and the power law slopes, β , are quite close for the two curves. Although the two curves differ by a factor of 2.5 in resistance, the estimates of β differ by 1% and those of K_{H_2} by 40%. Again we find that the sensitivity coefficient and power law slope vary little at a fixed temperature, they are peculiar to the particular detector and independent of ambient conditions. The constant K_0 of eqn (5) is included to account for small differences in the impurity contents of the two constituted ambients.

There are two dramatic qualities of the hydrogen response of Fig 5 and eqn (5). The first is the order of the response in which the hydrogen concentration term appears as a square. For high concentration of hydrogen eqn (5) may be expressed as

$$R^{1/\beta} \sim [O_2]/[H_2]^2$$

The resistance response is first order in oxygen pressure and second order in hydrogen pressure, perhaps implying a mass action relationship indicative of the stoichiometries of a chemical reaction. The second is the extreme sensitivity of the sensor to hydrogen. This arises from two factors. First, the second-order nature of the response produces a power law slope at high concentrations which is twice that of many other gases (*e.g.*, methane and water). Secondly, the gas sensitivity coefficient is more than two orders of magnitude greater for hydrogen than for methane. Only 10 ppm H_2 in air is sufficient to decrease the resistance by a factor of two. One per cent H_2 , still a factor of five below the lower explosive limit, reduces sensor resistance by three orders of magnitude.

Carbon monoxide low concentration response

The response of the TGS to carbon monoxide is also found to be a power law relationship conforming to an expression of similar form to those of methane, hydrogen or water. For concentrations below 3000 ppm, the carbon monoxide response is first order with respect to oxygen. The sensitivity coefficient for carbon monoxide depends greatly on the water vapor content of the ambient. This is illustrated by the results of the five separate measurements of the CO response displayed in Fig 6. The 100 ppm water

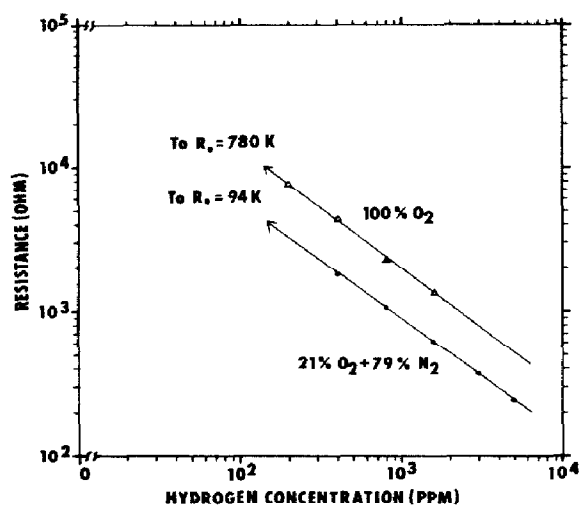


Fig 5 Resistance response of the TGS to hydrogen in different ambients. Points represent experimentally determined values and the solid lines are curve fits to eqn (5)

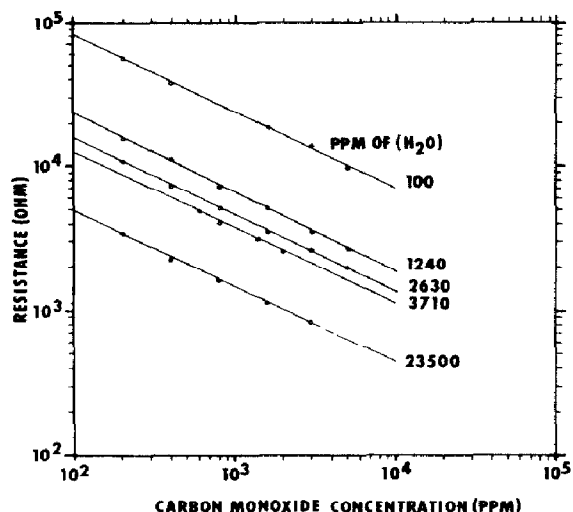


Fig 6 Resistance response of a single sensor to carbon monoxide in air with water concentration as a parameter. The response is first order with a sensitivity coefficient proportional to the water vapor concentration

concentration was the lowest concentration which could be reliably obtained while the highest, 23 500 ppm, resulted from complete saturation of the ambient air with water

Each response curve of Fig 6 exhibits the same power law slope, β . The sensitivity coefficients differ, however, indicating a dependence on the water vapor content. The five curves conform to the equation,

$$R = R_0(K'_{CO}[CO])^{-\beta} \quad (6)$$

in which K'_{CO} is related to water vapor concentration. Alternately, the TGS response to H_2O concentration in the presence of CO can be expressed as

$$R = R_0(K_{H_2O}[H_2O])^{-\beta} \quad (7)$$

where K_{H_2O} depends on CO concentration

The equality of eqns (6) and (7) suggests that the sensor resistance must depend upon the product of the water and carbon monoxide concentrations rather than the individual concentrations. It is found that the entire family of curves of Fig 6 is well represented by the single expression,

$$R = R_0 P_{O_2}^\beta (1 + K_{CO}[H_2O][CO])^{-\beta} \quad (8)$$

which describes the CO response for concentrations below 3000 ppm. For greater concentrations, the power law slope increases so that eqn (8) is no longer applicable

Carbon monoxide general response

In order to quantify the entire response of TGS resistance to carbon monoxide, from zero to very high concentrations, the experiment illustrated by Fig 7 has been performed. A sensor is immersed in a constituted air ambient containing 100 ppm H_2O . Its response to CO concentrations up to 30 000 ppm is measured (curve ABC). The sensor resistance is then recorded as the H_2O concentration is gradually increased to 1200 ppm (curve DE). Finally, the CO concentration is raised to 80 000 ppm while the H_2O concentration is held constant at 1200 ppm.

It can be noted from Figs 6 and 7 that the TGS response is first order in both CO and H_2O for low H_2O and low CO concentration (curve AB), second order in CO at high CO and low H_2O concentrations (curve BC), first order in H_2O at high CO and high H_2O concentrations (curve DE), and second order in CO at high CO and high H_2O concentrations (curve FG).

A single mathematical expression, which combines the first-order CO dependence for low concentrations and the second-order CO dependence for high concentrations takes the form

$$R = R_0 P_{O_2}^\beta (1 + K_{H_2O}[H_2O] + K_{CO}[H_2O][CO] + K'_{CO}[H_2O][CO]^2)^{-\beta} \quad (9)$$

in which $R_0 = 265 \text{ k}\Omega$, $K_{H_2O} = 3 \times 10^{-3} \text{ ppm}^{-1}$, $K_{CO} = 1.5 \times 10^{-3} \text{ ppm}^{-2}$, $K'_{CO} = 1.73 \times 10^{-7} \text{ ppm}^{-3}$, $P_{O_2} = 1$ for air, $\beta = 0.42$ and with $[H_2O]$ and $[CO]$ in ppm.

This single equation is represented by the solid lines of Figs 6 and 7. The response model approximates the measured resistances with an average error of 2.5%, which is comparable to the measurement precision. That eqn (9) accurately describes not only the regions of first- and second-order response but also the transition region connecting the two, strongly indicates the correctness of its form. Equation (9) also clearly shows the dependence of the sensitivity of the TGS to CO on H_2O concentration. Thus, in the low CO and H_2O concentration regime, the threshold of detection occurs when

$$[H_2O][CO] = K_{CO}^{-1}$$

For H_2O concentrations of the order of 1000 ppm, the threshold of CO detection is only 1 ppm. The transition between first- and second-order responses occurs at a CO concentration determined only by K_{CO} and K'_{CO} and independent of H_2O concentration,

$$[CO] = K_{CO}/K'_{CO}$$

Combined effect of O_2 , H_2O , CO, CH_4

To determine the influence of the simultaneous presence of various gases on the TGS detection properties, the response of the TGS resistance to carbon monoxide concentrations in different ambients is measured. One ambient consists of air with 100 ppm H_2O , the other is composed of oxygen containing 300 ppm H_2O and 3000 ppm CH_4 . The same sensor whose CO

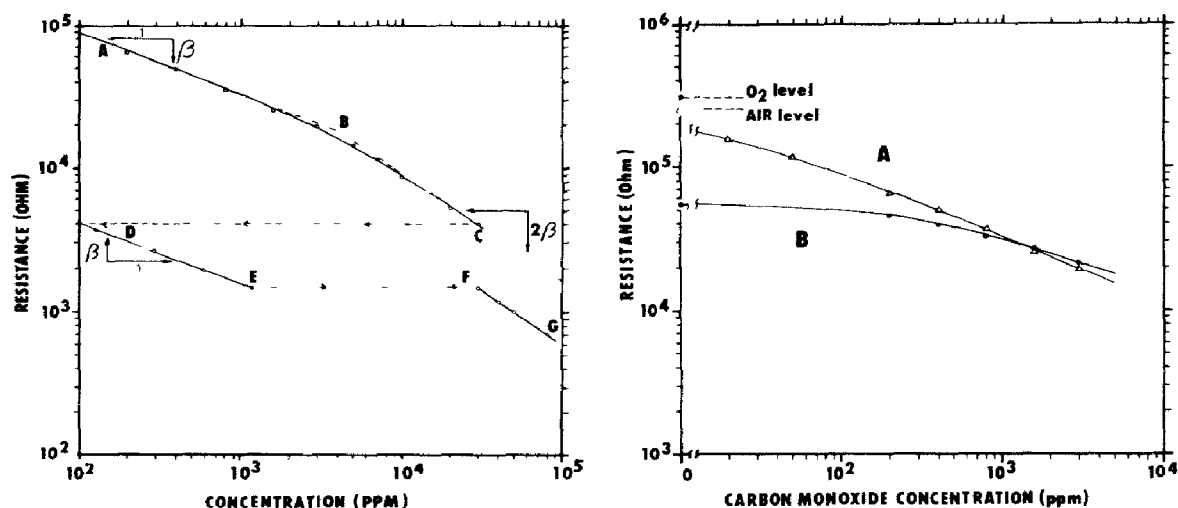


Fig 7 General resistance response of the TGS to CO and H₂O. Curve A–C, response to CO in air containing 100 ppm H₂O. Curve D–E, response to H₂O in air containing 30 000 ppm CO. Curve F–G, response to high CO concentrations with 1200 ppm H₂O. The solid lines represent eqn (9).

Fig 8 Two responses of a single gas sensor to CO. Response A is measured in air containing 100 ppm H₂O. Response B is measured in oxygen containing 300 ppm H₂O and 3000 ppm CH₄. The solid curves both represent eqn (10).

response is displayed in Fig 7 is also used to explore sensor response to CO concentrations under the two ambient conditions shown in Fig 8.

The response curve labeled A is measured in air with 100 ppm H₂O. The values measured conform to eqn (9). As expected, the response approaches a constant resistance for very low concentrations and a first-order power law for higher concentrations of CO. The very high concentrations regime, at which the second-order power law becomes dominant, is not included.

Curve B of Fig 8 represents the response to CO concentrations in a 100% O₂ ambient to which 300 ppm H₂O and 3000 ppm CH₄ are added. This response exhibits lower resistances than case A at zero concentration of CO, owing to the methane contamination, and higher resistances at large CO concentrations, resulting from the greater oxygen partial pressure. Consequently, the power law slope of curve B appears less than that of curve A. Even so, curve fits to the response model demonstrate that the asymptotic slopes are identical.

It is this kind of seemingly complex sensor response that has, until now, characterized the TGS as being unsuited for general purpose gas detection. However, by proposing a model for the sensor response consistent with our various measurements, we can explain the behavior exhibited in Fig 8. The CO response model, eqn (9), is supplemented with a term representing the methane sensitivity to produce

$$R = R_0 P_{O_2}^\beta (1 + K_{CH_4}[CH_4] + K_{H_2O}[H_2O] + K_{CO}[H_2O][CO] + K'_{CO}[H_2O][CO]^2)^{-\beta} \quad (10)$$

with $K_{\text{CH}_4} = 0.08 \text{ ppm}^{-1}$ and the remaining parameter values those of eqn (9). This model is represented by the two solid curves of Fig 8 with the appropriate ambient conditions of P_{O_2} , $[\text{H}_2\text{O}]$, and $[\text{CH}_4]$ inserted. Again, the measurement data fit eqn (10) with a sample error approximating the measurement precision.

Combined effects of CH_4 and H_2

By analogy with the previous results, the combined action of methane and hydrogen is expected to conform to the following formulation

$$R = R_0 P_{\text{O}_2}^\beta (1 + K_{\text{CH}_4}[\text{CH}_4] + K_{\text{H}_2}[\text{H}_2]^2)^{-\beta} \quad (11)$$

Indeed all results agree well with this equation. However, it is difficult to verify in that region where the individual effect of methane and hydrogen are comparable. This difficulty results from the huge disparity between sensitivity coefficients for methane and hydrogen — that of hydrogen is approximately two orders of magnitude greater than that of methane. This, coupled with the observation that the response to hydrogen is second order while the response to methane is first order, precludes the discernment of methane response in all but the very lowest concentrations of hydrogen. Only 10 ppm H_2 influences sensor resistance to the same degree as 10 000 ppm of CH_4 . Hydrogen responses in concentrations of methane up to 10 000 ppm have been measured and found identical to the response of Fig 5.

Even though the much greater sensitivity to hydrogen creates difficulty in confirming eqn (10) in that region of gas concentrations where the effects of the two gases are comparable, for the regions in which one of the gases dominates eqn (11) reduces to the simpler expressions (3) or (5), which have been extensively verified.

Effects of temperature on the TGS response parameters

Having derived an empirical expression for the TGS response to a reducing gas, we use it as a framework for understanding the effects of sensor operating temperature on the sensor resistance response.

In its simplest form, the static reducing gas response illustrated by that of methane,

$$R = R_0(1 + K_{\text{CH}_4}[\text{CH}_4])^{-\beta} \quad (12)$$

is characterized by three independent parameters: the power law exponent, β , the intrinsic clean air resistance, R_0 , and the sensitivity coefficient, K_{CH_4} . Our observations, as well as data supplied by Figaro, Inc., Fig 9, indicate that the response depends critically on sensor temperature.

The temperature dependence of the response to a particular gas may be represented as a variation with temperature of one or more of the parameters contained in eqn (12). To quantify the effect of sensor operating temperature on these parameters, careful measurements of the methane response in air were undertaken at various sensor temperatures for a single sensor.

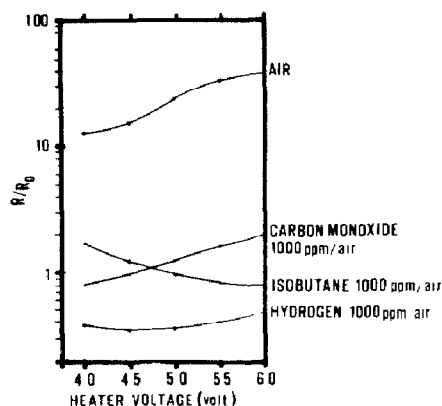


Fig 9 Effect of heater voltage on TGS resistances R/R_0 is the ratio of steady-state resistance to its resistance in air containing 1000 ppm isobutane at 5 heater volts (after Figaro, Inc)

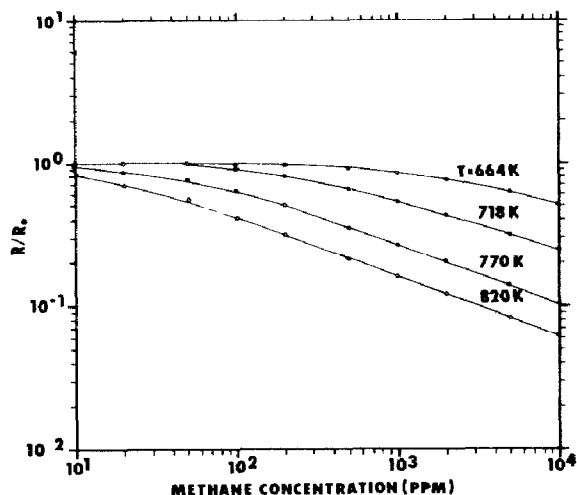


Fig 10 TGS resistance response to methane for different temperatures R/R_0 is the ratio of sensor resistance to the resistance at zero concentration for each temperature The solid curves represent eqn (12)

Methane was used because its response is simple, only first order, and it was already the gas studied in the greatest detail

The gas sensor tested was one for which the temperature was precisely measured by a technique described elsewhere [1] The temperatures corresponding to heater voltages were known to ± 4 K

It is important to note that for each temperature the sensor is allowed sufficient time to achieve steady state Generally this requires at least ten hours Only after this time are gas response measurements undertaken If this time is not allowed, a significant transient response to the temperature change obscures gas response measurements That transient response was the subject of extensive experimentation and is reported in the companion paper

The steady state methane response is measured six times, at sensor operating temperatures corresponding to heater voltages of 3.5, 4, 4.5, 5, 5.5 and 6 volts At the two lowest temperatures the methane sensitivity is too greatly diminished to provide a reliable measurement of the parameters of eqn (12) However, for each of the other responses the equation is fitted to the sensor response, and the parameters R_0 , β and K_{CH_4} are reliably determined

Figure 10 shows the sensor response to methane in air with sensor temperature as a parameter The ratio of resistance at each concentration to the resistance at zero methane concentration is plotted The effect of operating temperature on the responses of Fig 10 is dramatic The sensitivity increases rapidly with temperature while the power law slope appears to

change only slowly, the curves at high concentration are nearly parallel. Each of the response curves of Fig. 10 is independently fitted to eqn. (12) to determine best estimates of R_0 , β and K_{CH_4} . Again, the dots of Fig. 10 represent measured resistance ratios while the fit to the equation is indicated with a solid line.

The monotonic variation of R_0 with steady-state temperature is great and roughly linear with temperature. However, other observations indicate that this particular behavior extends only over a narrow temperature range.

Temperature variation of power law exponent

The variation of the power law exponent, β , with operating temperature is shown in Fig. 11. A least-squares linear regression yields

$$\beta = (-0.013 \pm 0.01) + (6.28 \pm 0.16)kT \quad (13)$$

where $k = 8.6171 \times 10^{-5}$ eV/K, and T in K. The errors are computed from the functional dependence of the determined parameter on the individual sample deviations from the best fit, assuming that β and T are precisely known.

It is evident that the constant term of eqn. (13) is zero within the measurement error, suggesting that β is directly proportional to absolute temperature. Figure 11 illustrates that proportionality; the line represents $\beta = 6.0 kT$. It should be emphasized that this determination is done for a single sensor; there is considerable intersensor variation ($\beta = 0.25$ to 0.55) at a particular temperature.

Temperature variation of gas sensitivity coefficient

The sensitivity coefficients for methane at each of the operating temperatures are shown in Fig. 12. The graph displays the logarithm of the gas sensitivity coefficient against inverse temperature. The straight line results from a linear least-squares fit of the data, and indicates an exponential dependence of K_{CH_4} on inverse temperature,

$$K_{CH_4} = K_{CH_4}^0 \exp[-E_a/kT] \quad (14)$$

The intercept at infinite temperature, $K_{CH_4}^0$, is 1.71×10^8 ppm⁻¹ and the activation energy E_a is 1.51 eV.

Discussion

Stability and reproducibility of the TGS response

The complex and highly non-linear response of the TGS is not amenable to the design of simple direct reading instrumentation. Some of its response idiosyncrasies have caused many researchers to consider it a non-reproducible sensor and, therefore, not of much promise for quantitative instrumentation. Thus whatever little there is in the literature on the TGS is of a qualitative nature [2 - 4].

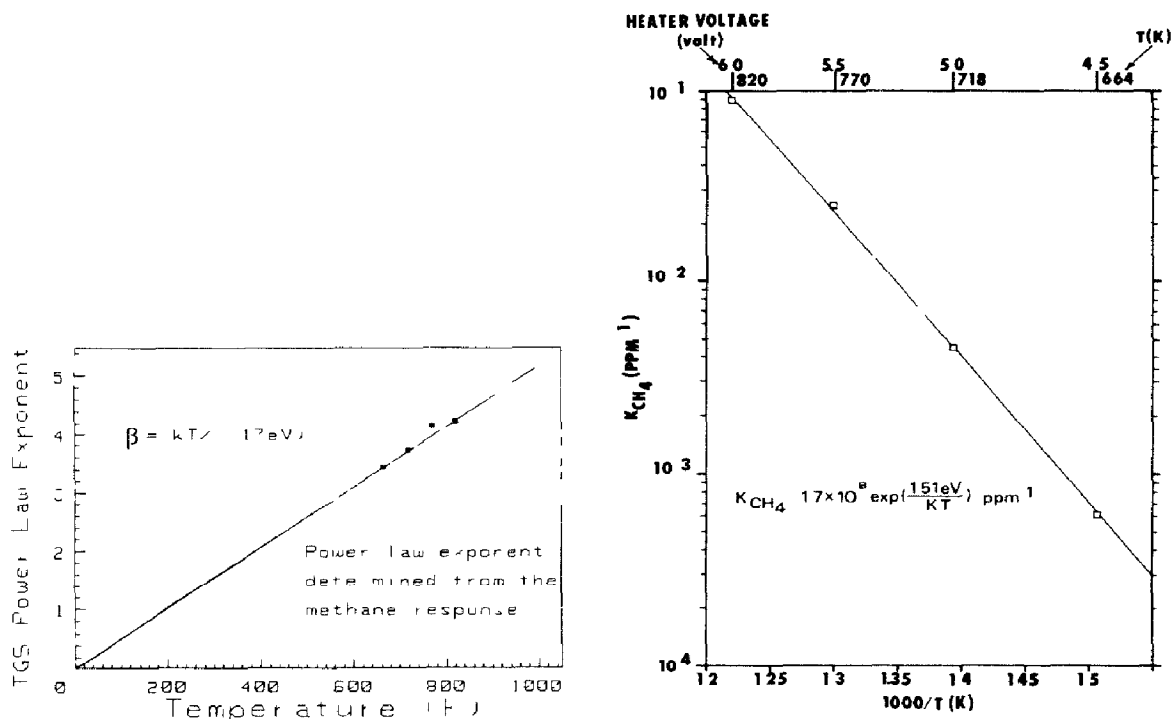


Fig 11 The power law exponents for the curves of Fig 10 are directly proportional to sensor operating temperature

Fig 12 The logarithms of the methane sensitivity coefficients are plotted as a function of reciprocal sensor temperature. The line, eqn (14), indicates an activated process characterized by an energy of 1.5 eV

The results of this study of the gas sensing characteristics of the TGS demonstrate long-term stability and reproducibility of gas response provided that measurements are made with careful regard to the conditions of contaminant gas concentrations and sensor operating temperature. For example, the sensitivity coefficient K_{CH_4} is obtained from the responses of the TGS to methane, measured numerous times over a period of months. For all these responses, the sensitivity coefficient remains the same even though the gas ambient is varied and the sensor had been previously exposed to diverse conditions of temperature and reducing gas concentrations.

In addition, the sensor gas response must be measured with sufficient detail over a very wide range of gas concentration in order for it to be fully characterized. The response data provided by the manufacturer are too scanty to provide even a reliable estimate of response order to each gas.

Sufficient care must also be exercised in quantifying the effects of the ambient gas when measuring the TGS resistance response to a particular gas. For instance, in Fig 8 the TGS responses to carbon monoxide under two very different ambient conditions of oxygen, water and methane concentrations are shown. Casual inspection of these responses suggests that the carbon monoxide response is irreproducible. However, they can be quanti-

tatively described by appropriate terms in eqn (10). When the effects of oxygen, water and methane concentrations are considered, each response can independently be used to determine the same value of sensitivity coefficient for carbon monoxide. Equation (10) provides the framework for understanding the contribution of different ambient conditions to this particular gas response.

Our results indicate long-term stability of the TGS gas responses. The ability to measure the same parameters of response irrespective of the long-term history of a sensor suggests the great metallurgical and chemical stability of the sensing material. This is in contrast to some other metal oxide materials, particularly zinc oxide, which we find undergo irreversible chemical change on long exposure to atmospheric air.

It must be emphasized that in this study response measurements are performed only after the sensor is operated at a fixed temperature for several days. This is necessary to obviate the transient and long-term drifts which result from temperature change. These kinetic variations have been extensively studied and are the subject of the companion paper.

Nature of the general response equation

We may generalize the results expressed in eqns (8) - (11) to provide an expression which describes the response of the TGS to any combination of reducing (combustible) gases

$$(R/R_0)^{-1/\beta} = \left(1 + \sum K_i [G_{1,i}]^{n_{1,i}} [G_{2,i}]^{n_{2,i}} \right) / [O_2] \quad (15)$$

where $[G]$ represents a reducing gas concentration and n is an integer or fractional integer power. For many terms of the summation there is only one term per gas, e.g., $[CH_4]$ or $[H_2O]$, for others there is a product of several, e.g., $[H_2O][CO]^2$.

The results of all measurements performed using O_2 , CH_4 , CO , H_2O and H_2 are consistent with eqn (15). The following significant terms in the summation have been identified: $K_{CH_4}[CH_4]$, $K_{H_2O}[H_2O]$, $K_{H_2}[H_2]^2$, $K_{CO}[H_2O][CO]$, and $K'_{CO}[H_2O][CO]^2$.

In the light of the measured responses of the previous section, various aspects of the general response equation are now discussed.

Power law

An important quality of the response is its power law nature, represented by the power law exponent, β . We find that the value of β varies between sensors, with the newer TGS sensors having greater values and a smaller intersensor variation. Previous investigators [4] have attempted to ascribe this seemingly arbitrary exponent to constant fixed-order reaction rates. This interpretation, however, does not account for the well-defined functional dependence of β on sensor temperature, or for its intersensor variation. The physical mechanisms responsible for a power law response exhibiting a non-integer exponent, β , and the physical and chemical pro-

cesses that cause the variation in β among sensors will be treated in a future paper

Role of oxygen

The general response expression suggests that oxygen plays the central role in the detection of combustible gases by the TGS. The effects of oxygen and all reducing gases are reciprocal,

$$(R/R_0)^{-1/\beta} \sim [G]^n/[O_2]$$

for any gas in great concentration. Assuming that some ionosorbed oxygen species controls the sensor resistance, then oxygen pressure increases the population of that species while reducing gases, through oxidation-reduction reactions, decrease it. In steady state we would expect populations to be determined from a mass action expression containing integer or fractional integer powers of oxygen and reducing gas pressures.

To determine β properly, the sensor response to O_2 needs to be measured. This obviates the difficulty of simultaneously determining both sensor specific β and the order of the reducing gas effect by measuring only the reducing gas response.

Sensitivity

The general response, eqn (15), provides a rigorous definition of sensitivity. The K_j s may be termed sensitivity coefficients and are determined from the measured response. This sensitivity definition allows meaningful comparisons of sensor performances, and provides the measurement basis for further detector research. Because this type of response appears to be generally applicable to metal oxide sensors, the K_j s may be used to compare the sensitivity of various types of gas sensor materials and to provide an absolute measure of the catalytic activity of a sensor's surface.

The detection threshold — that concentration of reducing gas necessary for a meaningful change in resistance — depends not only on absolute sensitivity to that particular gas but also on concentrations of other gases in the ambient which partially mask the response to the gas of interest. This behavior, evident in eqn (15), is one source of many inconclusive, irreproducible or contradictory measurements of what investigators have called 'gas sensitivity' but which may have been a change in the detection threshold. If the appropriate simplification of eqn (15) is used to fit the response data, then many responses with different detection thresholds will yield reproducible gas sensitivity coefficients, alleviating much confusion.

There is a difficulty in measuring the gas sensitivity coefficient when using an ambient gas with unknown impurities. To illustrate this difficulty, consider the general response of eqn (15) (excluding for convenience the O_2 term) to methane gas

$$R^{-1/\beta} = R_0^{-1/\beta} \left(1 + \sum K_j [G_{1j}]^{n_{1j}} \right) + R_0^{-1/\beta} K_{CH_4} [CH_4] \quad (16)$$

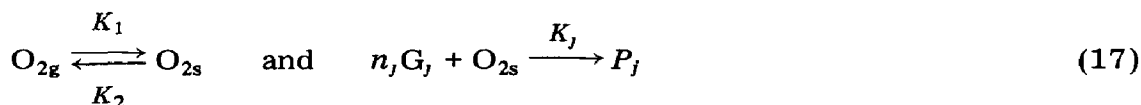
in which the $[\text{CH}_4]$ term has been isolated from the sum. From a measurement of the sensor resistance at many methane concentrations, three parameters may be determined: β , the entire first term of the right side of eqn (16) and the product $R_0^{-1/\beta} K_{\text{CH}_4}$. Unless the ambient used (generally air) is very pure, the summation term is large with respect to one and the first term of the right side of the above equation does not simply represent $R_0^{-1/\beta}$. Consequently R_0 cannot be separated from the $R_0^{-1/\beta} K_{\text{CH}_4}$ term. K_{CH_4} is not determined, but only the product $R_0^{-1/\beta} K_{\text{CH}_4}$. Because R_0 varies from one sensor to another, precise comparisons of sensitivity coefficient *between* sensors become difficult.

There are two means of remedying this problem. The first is to use pure constituted ambient air. The second is to measure the response to several reducing gases in the same ambient. The sensitivity to a single gas is not absolutely known, but precise measurements of the ratios of sensitivity coefficients for two gases can be attained. The ratio of the two parameters determined in eqn (16) then gives this ratio directly, irrespective of the value of R_0 .

Combinations of gases

The response model, eqn (15), provides a cogent mathematical description for the qualitatively observed *competitive* and *synergistic* interactions among gases. Because the resistance response depends on a linear combination of individual gas terms, the effects of one gas can be masked by the combined effects of others. A single gas's influence cannot be discerned until its contribution to the sum of eqn (15) becomes significant. In addition to this competition, there is an associative interaction by which the effects of one gas are enhanced by the presence of another. This synergism occurs when the product of several gas concentrations constitutes a single term in the sum.

A heuristic argument is presented here to demonstrate physical plausibility for the form of eqn (15). A more detailed physical theory of the source of eqn (15) will be presented in a future paper. The addition of gas effects in eqn (15) may be understood if it is assumed that each term represents a separate reaction for the removal of adsorbed surface oxygen on the TGS. Consider these reactions



where O_{2g} is gaseous oxygen and O_{2s} surface adsorbed oxygen, K_1 and K_2 are the exchange rates for oxygen between the gaseous and adsorbed states, respectively, G_j is a reducing gas chemically reacting with surface oxygen at a rate K_j , yielding a product P_j that is desorbed from the surface. For each reducing gas there is a different surface reduction equation and reaction rate. The rate equation for the O_{2s} concentration is written as

$$\frac{d}{dt}[\text{O}_{2s}] = K_1[\text{O}_{2g}] - K_2[\text{O}_{2s}] - \sum K_j[\text{O}_{2s}][G_j]^{n_j} \quad (18)$$

If it is stipulated that the sensor resistance depends on the O_{2s} concentration raised to a power,

$$R/R_0 \sim [\text{O}_{2s}]^\beta \quad (19)$$

then in the steady state (*i.e.*, $d/dt = 0$) the sensor resistance is given by,

$$(R/R_0)^{-1/\beta} = \left(K_2 + \sum K_j[G_j]^{n_j} \right) / K_1[\text{O}_{2g}] \quad (20)$$

Realizing that a combination of reducing gases sometimes participates in a single reduction reaction, we obtain the equivalent of eqn (15). Each independent reaction sequence contributes a term to the summation. Comparing eqn (20) to the empirically determined response eqn (15), we identify the gas sensitivity coefficients with reaction rates.

This interpretation is substantiated by the finding that the K_j exhibit an exponential dependence on reciprocal temperature, eqn (14). The measured activation energy for the methane reaction, $E_a = 1.5$ eV, is typical for the catalytic oxidation of hydrocarbons over metal oxides [6-9]. Justification for the assumption, eqn (19), and a description of the origin of the power law exponent, β , based on the pertinent chemical and semiconductor theory will be presented in a future paper.

Particular gas response terms

$$K_{\text{CH}_4}[\text{CH}_4]$$

The first-order response to methane is the most carefully measured response of this investigation. All older (*c.* 1976-1979) TGS detectors exhibit the single well-defined first-order responses of Figs 2 and 3. Preliminary measurements of the responses of some more recent samples show signs of a second-order response to methane in addition to the first-order response. This effect has not yet been thoroughly investigated.

$$K_{\text{CO}}[\text{H}_2\text{O}][\text{CO}] \text{ and } K'_{\text{CO}}[\text{H}_2\text{O}][\text{CO}]^2$$

The influence of water vapor in the ambient on the sensor response to CO had previously been qualitatively observed [2]. Here we have quantified the effect, measured the response order and determined that there are at least two reactions of importance in the TGS response to carbon monoxide. Both rely on the presence of water, are first order in their sensitivity to water, and remove surface oxygen.

That CO reacts with adsorbed oxygen has been verified on zinc oxide [10] and that hydroxyl groups may play a role has been noted by Thornton and Harrison [5] in infrared spectrographic studies of SnO_2 . Further investigations, particularly of the temperature dependences of the sensitivity coefficients, may produce activation energies of the reactions.

Thornton and Harrison [5] and Boccuzzi *et al* [11] have determined that CO is adsorbed adjacent to a hydroxyl group on both SnO₂ and ZnO. If so, a hydroxyl group of origin other than water adsorption may also promote the detection of CO. It is likely that hydrogen adsorption might produce just such a site by reaction with surface oxygen. In that case, we might expect another term of the CO response $K_{CO}[H_2][CO]$. The independent sensitivity of the TGS to hydrogen makes detection of such a term difficult. Nevertheless the search for such a term may be a fruitful avenue for further investigation.

$$K_{H_2O}[H_2O]$$

The first-order sensitivity of the TGS to water vapor is of great chemical interest, as H₂O is not generally considered a reductant. Consequently the existence of an independent sensitivity to water, separate from the water-related CO response, is of great value in contributing to knowledge of the catalytic processes involved in detection. It is possible that the water response results from the presence of CO or some other contaminant (probably polar) in the ambient. Comparing the measured value of K_{H_2O} to that of $K_{CO}[CO]$, we find that of the order of 10 ppm CO contamination would cause the response ascribed to water. That is a quantity much greater than the CO concentration that can be present in the ambient gas prepared in our laboratory, making it likely that the H₂O response is independent of other gases. Further investigation of the H₂O response is necessary to search for other gas interactions and to measure the water sensitivity coefficient accurately.

The practical importance of this response results from the substantial and variable water concentrations in atmospheric air.

$$K_{H_2}[H_2]^2$$

The TGS response to hydrogen is second order and independent of the other gases investigated. Because reactions of hydrogen and oxygen are perhaps the best understood of oxidation-reduction reactions, further investigation of this term might provide insight into the nature of reduction reactions on the TGS surface.

Determination of reaction order

The most significant characteristic of catalysis on metal oxide surfaces is the observation of the power law of non-integer exponent. Previous attempts to relate these power laws to reaction rates or mass action laws of fixed order have not been successful [12].

Non-integer order reaction rates appear to be a general characteristic of SnO₂ [13] and other metal oxides [12, 14]. A thorough study [6 - 9] of the oxidation reactions of hydrocarbons and CO over metal oxide catalysts in which the rate of product (CO₂) generation is measured has shown non-integer and non-fractional order kinetics to be the dominant feature of the reaction. In that study of NiO, Cr₂O₃ and Co₃O₄ catalyst beds, the rate of

product production is found to depend, in all cases, on non-integer powers of oxygen and hydrocarbon/CO partial pressures

$$d/dt[\text{CO}_2] = K(P_{\text{O}_2}^m P_{\text{HC/CO}}^n)/P_{\text{H}_2\text{O}}^l \quad (21)$$

The investigators find that CO and hydrocarbons are stoichiometrically converted to CO_2 and H_2O with reaction orders l, m, n in the range 0.3 - 0.5 for a Co_3O_4 catalyst. This is the same range we observe for the TGS power law exponent.

Additionally, we have found that for the TGS, embedded within the seemingly arbitrary power law response, careful measurements reveal integer-order responses which may be indicative of reaction stoichiometries. The superposition of the power law exponent, β , on the summation of eqn (15) greatly complicates empirical determinations of reaction orders. Nevertheless, it is possible to deduce reaction stoichiometries from the response data by performing detailed measurement of the response and curve fitting with the appropriate limiting form of eqn (15).

Conclusions

The static gas responses of the Taguchi Gas Sensor to methane, hydrogen, water, carbon monoxide and oxygen have been characterized in detail. The results indicate that when measurements are performed with stable and well-defined ambient conditions and operating temperatures, the TGS exhibits a long-term stable and reproducible response characterized by parameters which can be reliably measured.

Detailed measurement of the TGS resistance response to individual gases as well as to combinations of gases resulted in an empirically derived general response, eqn (15), which accurately describes the response of the TGS under all static gas conditions investigated. Some physical arguments for the form of the equation are presented, and its general properties are discussed.

Furthermore, from the experimental results and the formulation of the empirical relation, we note the following:

1. A precise mathematical definition of sensitivity of the TGS to various gases has been formulated which allows meaningful quantitative comparisons of sensor performances.

2. The relative sensitivities of the TGS to various reducing gases can be determined in the presence of other unknown contaminating gases by fitting the measured response to the appropriate form of the mathematical model. The detection threshold for a particular gas depends on the concentrations and kinds of other contaminating gases.

3. The mathematical formulation provides for the competitive and synergistic interactions of several gases detected simultaneously.

4. Oxygen gas plays an important complementary role to reducing gases in the detection process of the TGS. A model which assumes that sensor

resistance is determined solely by the surface concentration of adsorbed oxygen is sufficient to account for all aspects of the measured combustible gas response

5 The response of the TGS to carbon monoxide is greatly influenced by the simultaneous presence of water vapor. This interdependence is rigorously quantified and two reaction terms of differing stoichiometries are identified

6 The form of the empirical relation that predicts the TGS response to various gases in the ambient is suggestive of the possible catalytic activities occurring on the TGS surface

7 The non-integer power law exponents evident in a sensor's resistance response to reducing gas concentrations may be resolved as the product of two terms. The first term cannot be expressed as the ratio of integers, it is specific to the individual sensor, it is directly proportional to absolute temperature and it is independent of the gas type. The second term is an integer which is constant with respect to temperature, it is specific to the gas type, and it is the same for each sensor. These observations are consistent with the hypothesis that the second term represents the stoichiometry of the pertinent oxidation-reduction reaction. Both this term and the reaction activation energy can be derived from a sensor's resistance response by curve fitting to the mathematical model

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Biographies

Paul K Clifford is a Research Scientist for the Robotics Institute of Carnegie-Mellon University. He received a B S in Physics and Electrical Engineering in 1976 and an MS in Measurement and Control Science in 1977 at Carnegie-Mellon University. He continued his research at CMU as a Schlumberger Fellow and was awarded a Ph D in Electrical Engineering in 1981. His doctoral research consisted of a fundamental experimental and theoretical investigation of semiconductor materials used as sensing materials for flammable, toxic and pollutant gases.

From January, 1982 to May, 1982 he was a Visiting Lecturer in Electrical Engineering at CMU and is now a Research Scientist for the Robotics Institute. His present interests include the development of microfabricated mechanical and chemical sensors, the development of 'smart' robotic sensors through the integration of on-chip signal processing, gas detection and electrocatalysis at semiconductor electrodes.

David T Tuma is Professor of Electrical Engineering at Carnegie-Mellon University. He received his B S degree in Engineering from the American University of Beirut in 1961, and the Ph D degree in Electrical Engineering from the University of California at Berkeley in 1966. He has held academic positions at the University of California at Berkeley and Irvine. His research interests are in plasma electronics, solid state gas sensors, computer-based instrumentation and simulation and applied electromagnetic fields. Professor Tuma is a senior member of IEEE, and a member of the American Physical Society, the American Society for Engineering Education, Sigma Xi, the Spectroscopy Society of Pittsburgh and the Current Zero Club.